

# WI4014TU – Numerical Analysis for PDE's

## Lecture 8

### Finite Element Method

Mathematical preliminaries: function spaces, differential operators, functionals, variational calculus

## Function spaces

# Function Space

- $\Omega$  – *open* and *bounded* domain in  $\mathbb{R}^n$ , where “open” means that the set does not contain its boundary
- $\overline{\Omega}$  – *closure* of  $\Omega$ , union of  $\Omega$  and its boundary  $\partial\Omega$ , i.e.,  $\overline{\Omega} = \Omega \cup \partial\Omega$
- $V$  – *vector space* over *field*  $F$ . For every  $\mathbf{v}, \mathbf{u}, \mathbf{w} \in V$  and any  $a, b \in F$ :
  - 1 Associativity of addition:  $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
  - 2 Commutativity of addition:  $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
  - 3 Identity element of addition: there is  $\mathbf{0} \in V$  such that  $\mathbf{v} + \mathbf{0} = \mathbf{v}$
  - 4 Inverse element of addition: there is  $-\mathbf{v} \in V$  such that  $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$
  - 5 Compatibility of scalar and field multiplications:  $a(b\mathbf{v}) = (ab)\mathbf{v}$
  - 6 Identity element of scalar multiplication:  $1\mathbf{v} = \mathbf{v}$ , where  $1 \in F$
  - 7 Distributivity of scalar multiplication (1):  $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
  - 8 Distributivity of scalar multiplication (2):  $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$
- $V(\Omega)$  – *function space*. Given  $\Omega \subset \mathbb{R}^n$ ,  $V(\Omega)$  is a space of functions on  $\Omega$  (by default – vector space) that satisfy some smoothness requirements and possibly boundary conditions on  $\partial\Omega$

# Normed Space

- Assume  $F = \mathbb{R}$
- $\|\mathbf{v}\|$  – *norm* of  $\mathbf{v} \in V(\Omega)$ , is an operation  $V(\Omega) \rightarrow F$  that “measures” the magnitude of the elements of  $V(\Omega)$ . For any  $\mathbf{u}, \mathbf{v} \in V(\Omega)$  and  $a \in F$  a norm must be:
  - ① Absolutely homogeneous (scalable):  $\|a\mathbf{v}\| = |a| \|\mathbf{v}\|$
  - ② Subadditive (triangle inequality):  $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$
  - ③ Nonnegative:  $\|\mathbf{v}\| \geq 0$
  - ④ Definite (point separating): if  $\|\mathbf{v}\| = 0$ , then  $\mathbf{v} = \mathbf{0}$
- *Normed vector space* is a pair  $(V, \|\cdot\|)$ , where  $V$  is a vector space and  $\|\cdot\|$  is a norm on  $V$
- $(V(\Omega), \|\cdot\|)$  – *normed (function) space*

## $L^2(\Omega)$ space

- Let  $\Omega = (a, b) \subset \mathbb{R}$
- $L^2(\Omega)$  – space of *square-integrable* functions, i.e.,

$$u \in L^2(\Omega) \text{ iff } \int_a^b |u(x)|^2 dx < \infty$$

- $L^2(\Omega)$  is a normed space with  $L^2$ -norm defined as

$$\|u\|_2 = \left( \int_a^b |u(x)|^2 dx \right)^{1/2}$$

- *Inner product* of  $u(x)$  and  $v(x)$  can be defined as:

$$(u, v)_{L^2} = \langle u, v \rangle_{L^2} = \int_a^b u(x)v(x) dx$$

- *Hilbert space* is a normed space, where norm is induced by the inner product.  $L^2(\Omega)$  is a Hilbert space, since  $\|u\|_2^2 = (u, u)_{L^2}$

## $H^1(\Omega)$ space

- $H^1(\Omega)$  – Sobolev space of square-integrable functions with square-integrable first derivatives, i.e.,

$$u \in H^1(\Omega) \text{ iff } u \in L^2(\Omega) \text{ and } \frac{du}{dx} \in L^2(\Omega)$$

Hence,  $H^1(\Omega) \subset L^2(\Omega)$

- $H^1(\Omega)$  is a normed space with the norm defined as:

$$\|u\|_{H^1} = \left( \int_a^b \left[ |u(x)|^2 + \left| \frac{du}{dx} \right|^2 \right] dx \right)^{1/2}$$

- $H^1(\Omega)$  is a Hilbert space since its norm is induced by an inner product:

$$(u, v)_{H^1} = (u, v)_{L^2} + \left( \frac{du}{dx}, \frac{dv}{dx} \right)_{L^2} \Rightarrow \|u\|_{H^1}^2 = (u, u)_{H^1}$$

## $C^k(\Omega)$ spaces

- $C(\Omega)$  – space of functions *continuous* on  $\overline{\Omega}$
- $C(\Omega) \subset L^2(\Omega)$ , since if  $u \in C(\Omega)$  and  $\overline{\Omega}$  is a *compact* set, then  $u$  is *bounded* and, therefore, square-integrable
- With the *uniform* norm:

$$\|u\|_{\infty} = \max_{x \in \overline{\Omega}} |u(x)|$$

$C(\Omega)$  becomes a normed space, but *not* a Hilbert space

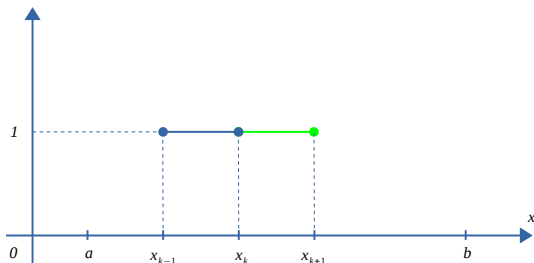
- $C^1(\Omega)$  – space of functions with *continuous first derivatives* on  $\overline{\Omega}$
- $C^2(\Omega)$  – space of functions with *continuous second derivatives* on  $\overline{\Omega}$

# Piecewise-constant functions

$$\Omega = (a, b)$$

$$c_k(x) = \begin{cases} 1 & \text{if } x \in [x_{k-1}, x_k] \\ 0 & \text{otherwise} \end{cases}$$

$$c_k(x) \in L^2(\Omega), c_k(x) \notin C(\Omega), c_k(x) \notin H^1(\Omega)$$

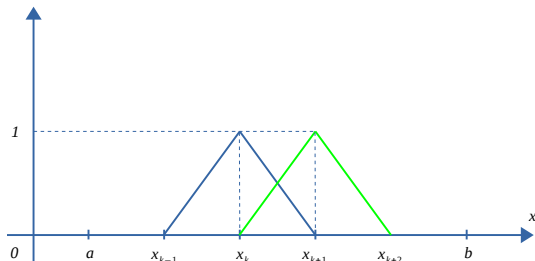




## Linear functions: $c_0 + c_1x$

$$\text{per node: } \ell_k(x) = \begin{cases} \ell_k^-(x) = \frac{x - x_{k-1}}{x_k - x_{k-1}} & \text{if } x_{k-1} \leq x \leq x_k \\ \ell_k^+(x) = \frac{x_{k+1} - x}{x_{k+1} - x_k} & \text{if } x_k \leq x \leq x_{k+1} \\ 0 & \text{otherwise} \end{cases}$$

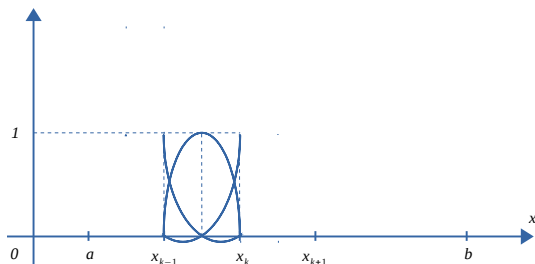
- $\ell_k(x) \in C(\Omega)$ ,  $\ell_k(x) \in H^1(\Omega)$ ,  $\ell_k(x) \in L^2(\Omega)$
- $\ell_k(x) \notin C^1(\Omega)$



## Quadratic polynomial functions: $c_0 + c_1x + c_2x^2$

per element  $x \in [x_{k-1}, x_k] : v(x) = \begin{cases} v_1(x) = \ell_{k-1}^+(2\ell_{k-1}^+ - 1) \\ v_2(x) = \ell_k^-(2\ell_k^- - 1) \\ v_3(x) = 4\ell_{k-1}^+\ell_k^- \end{cases}$

- $v(x) \in C(\Omega)$ ,  $v(x) \in H^1(\Omega)$ ,  $v(x) \in L^2(\Omega)$
- $v(x) \notin C^1(\Omega)$



# Function spaces and boundary conditions

- Functions from  $C(\Omega)$  satisfying homogeneous Dirichlet BC's:

$$C_0(\Omega) = \{u \in C(\Omega) \mid u(a) = u(b) = 0\}$$

- Similarly, we can define  $C_0^1(\Omega)$  and  $C_0^2(\Omega)$
- Let  $\Omega \subset \mathbb{R}^2$ , and  $\partial\Omega = \partial\Omega_D \cup \partial\Omega_N$ , where  $u(\mathbf{x}) = 0$  if  $\mathbf{x} \in \partial\Omega_D$ . If we say that  $u(\mathbf{x}) \in C_0^1(\Omega)$ , then it means

$$u(\mathbf{x}) \in C_0^1(\Omega) = \{u \in C^1(\Omega) \mid u(\mathbf{x}) = 0, \mathbf{x} \in \partial\Omega_D\}$$

Note that  $\partial\Omega_N$  – Neumann boundary – is not included in the definition of the function space  $C_0^1(\Omega)$

# Function spaces and boundary conditions

- A function space  $V_0(\Omega) = \{u \in V(\Omega) \mid u(x) = 0, x \in \partial\Omega\}$  with zero Dirichlet boundary conditions is often closed under addition and scalar multiplication, i.e., for any  $a, b \in \mathbb{R}$  and  $u, v \in V_0(\Omega)$ , we have  $au(x) + bv(x) \in V_0(\Omega)$
- A function space  $V_g(\Omega) = \{u \in V(\Omega) \mid u(x) = g(x), x \in \partial\Omega\}$  with nonzero Dirichlet boundary conditions is not closed under addition and scalar multiplication, i.e., if  $u, v \in V_g(\Omega)$ , then, in general,  $au(x) + bv(x) \notin V_g(\Omega)$

## Differential operators

# Differential operators

- To define a *differential operator* one needs a pair  $(D, V(\Omega))$ , where  $D$  is an *operation* of differentiation and  $V(\Omega)$  is a *function space*. For example, the operator of the second derivative on a function space of twice continuously differentiable functions on  $\Omega = (a, b)$  satisfying the BC  $u(a) = u(b) = 0$  is defined as

$$\mathcal{L}(u) = \left\{ \frac{d^2 u}{dx^2} \mid u \in C_0^2(\Omega) \right\}$$

- The  $n$ -dimensional linear differential operator  $\mathcal{L}$  of the order  $m$  is a *mapping* (*map*) between two function spaces  $\mathcal{L} : V(\Omega) \rightarrow W(\Omega)$ ;  $\mathcal{L}(u) = f$ , where  $u \in V(\Omega)$  and  $f \in W(\Omega)$ , such that

$$\mathcal{L}(u) = \sum_{|\alpha| \leq m} a_\alpha(x_1, x_2, \dots, x_n) \frac{\partial^{|\alpha|} u}{\partial x_1^{\alpha_1} \partial x_2^{\alpha_2} \dots \partial x_n^{\alpha_n}},$$

where  $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)$  is a multi-index and  $|\alpha| = \alpha_1 + \alpha_2 + \dots + \alpha_n$ . For example, a two-dimensional second-order differential operator is a mapping of the form:

$$\mathcal{L}(u) = a_{1,1}(x_1, x_2) \frac{\partial^2 u}{\partial x_1^2} + a_{1,2}(x_1, x_2) \frac{\partial^2 u}{\partial x_1 \partial x_2} + a_{2,2}(x_1, x_2) \frac{\partial^2 u}{\partial x_2^2}$$

# The role of function space

- Properties of  $\mathcal{L}$  depend on the choice of  $V(\Omega)$
- Dirichlet boundary conditions at  $\partial\Omega_D$  are included in the definition of the operator  $\mathcal{L}$  via the function space  $V(\Omega)$ , e.g., as  $V(\Omega) = C_0^1(\Omega)$

# Positive differential operators

Similar to positive-definite matrices, a differential operator  $\mathcal{L}$  is *positive* if

$$\int_{\Omega} v \mathcal{L}(v) d\Omega > 0, \quad \text{for any } v \in V(\Omega) \setminus \{0\}$$

where  $\{0\}$  is the zero element of the function space, e.g.,  $v(x) = 0, x \in \Omega$



# Positivity of Laplacian with Dirichlet BC's in 1D

The operator  $\mathcal{L}(u) = -u''$  is positive on  $C_0^1(a, b)$ :

$$\int_{\Omega} v \mathcal{L}(v) d\Omega = - \int_a^b v(x) v''(x) dx = \begin{bmatrix} (vv')' = v'v' + vv'' \\ vv'' = (vv')' - v'v' \\ \int_a^b (vv')' dx = [vv']_a^b \end{bmatrix}$$

$$= - [v(x)v'(x)]_a^b + \int_a^b v'v' dx = \int_a^b |v'(x)|^2 dx > 0,$$

since  $v \in C_0^1(a, b)$  i.e.  $v(x)$  continuous and  $v(a) = v(b) = 0$ .

**Question:** can it be that  $v(x) = \text{const} \neq 0$ , i.e.  $v'(x) = 0$ ,  $x \in (a, b)$ , and  $v(a) = v(b) = 0$ ?

# Positivity of Laplacian with Dirichlet BC's in 2D and 3D

The operator  $\mathcal{L}(u) = -\nabla \cdot (k \nabla u)$  is positive on  $C_0^1(\Omega)$ :

$$\begin{aligned} \int_{\Omega} v \mathcal{L}(v) d\Omega &= - \int_{\Omega} v \nabla \cdot (k \nabla v) d\Omega \\ &= \left[ \begin{aligned} \nabla \cdot (v k \nabla v) &= (\nabla v) \cdot (k \nabla v) + v \nabla \cdot (k \nabla v) \\ v \nabla \cdot (k \nabla v) &= \nabla \cdot (v k \nabla v) - (\nabla v) \cdot (k \nabla v) \end{aligned} \right] \\ &\quad \int_{\Omega} \nabla \cdot \mathbf{u} d\Omega = \int_{\partial\Omega} \mathbf{u} \cdot \mathbf{n} d\Gamma \\ &= - \int_{\partial\Omega} (v k \nabla v) \cdot \mathbf{n} d\Gamma + \int_{\Omega} (\nabla v) \cdot (k \nabla v) d\Omega \\ &= \int_{\Omega} k |\nabla v|^2 d\Omega > 0, \quad \boxed{\text{if } k(\mathbf{x}) > 0 \text{ and } v(\mathbf{x}) \in C_0^1(\Omega)} \end{aligned}$$

Specifically, we need  $k(\mathbf{x}) > 0$  for all  $\mathbf{x} \in \Omega$ ; and  $v(\mathbf{x}) = 0$ , for all  $\mathbf{x} \in \partial\Omega$

# Self-adjoint differential operators

The differential operator  $\mathcal{L}$  is *self-adjoint* on  $V(\Omega)$  if

$$\int_{\Omega} v \mathcal{L}(u) d\Omega = \int_{\Omega} u \mathcal{L}(v) d\Omega, \quad \text{for all } u, v \in V(\Omega)$$

This is the analogue of the matrix symmetry property.

# Self-adjointness of Laplacian with Dirichlet BC's in 1D

The operator  $\mathcal{L}(u) = -u''$  is self-adjoint on  $C_0^1(a, b)$ . Let  $u, v \in C_0^1(a, b)$ . Then,

$$\begin{aligned}\int_{\Omega} v \mathcal{L}(u) d\Omega &= - \int_a^b v(x) u''(x) dx \\ &= - [v(x) u'(x)]_a^b + \int_a^b v' u' dx = \int_a^b v' u' dx \\ \int_{\Omega} u \mathcal{L}(v) d\Omega &= - \int_a^b u(x) v''(x) dx \\ &= - [u(x) v'(x)]_a^b + \int_a^b u' v' dx = \int_a^b v' u' dx\end{aligned}$$

# Self-adjointness of Laplacian with Dirichlet BC's in 2D

**Theorem:** The operator  $\mathcal{L}(u) = -\nabla \cdot (k \nabla u)$  is self-adjoint on  $C_0^1(\Omega)$ , where  $\Omega \subset \mathbb{R}^2$

**Proof...**

## Functionals and variational calculus

# Functionals

- *Functional* is a mapping from a function space into the associated field of numbers. In the present case,  $L : V(\Omega) \rightarrow \mathbb{R}$
- It can be any mathematical operation that takes a function from  $V(\Omega)$  and produces a real number, e.g., integration or taking a maximum

## Example: distance on a plane

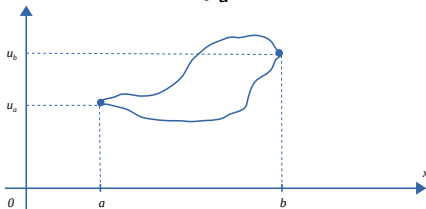
- Let  $V(\Omega)$  be  $C(a, b)$  with BC's  $u(a) = u_a$  and  $u(b) = u_b$  for each  $u \in V(\Omega)$ . In short

$$V(\Omega) = V(a, b) = \{u(x) \in C(a, b) \mid u(a) = u_a, u(b) = u_b\}$$

Then, every  $u \in V(\Omega)$  is a path between the points  $(a, u_a)$  and  $(b, u_b)$  in  $\mathbb{R}^2$ . The function space  $V(\Omega)$  is not closed (why?).

- Arc-length of each path is a functional

$$L(u) = \int_a^b \sqrt{1 + (u')^2} dx$$





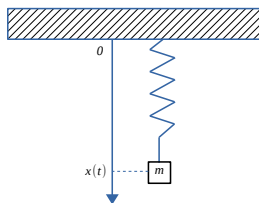
## Example: mass on a spring

- We consider  $x(t)$  from  $C(\Omega)$ , where  $\Omega = (a, b)$  is an open time interval, and BC's are  $x(a) = x_a$ ,  $x(b) = x_b$

$$V(\Omega) = V(a, b) = \{x(t) \in C(a, b) \mid x(a) = x_a, x(b) = x_b\}$$

- Kinetic energy is  $T = mv^2/2 = m(x')^2/2$ . Potential energy is  $U = kx^2/2$ , where  $k$  is the spring coefficient
- *Lagrangian* is  $\mathcal{L}(x', x, t) = T - U = m(x')^2/2 - kx^2/2$
- The *action* functional  $S$  is

$$S = \int_{t_1}^{t_2} \mathcal{L}(x, x', t) dt = \int_a^b \left( \frac{m(x')^2}{2} - \frac{kx^2}{2} \right) dt$$



x

# Minimization of functionals

- *Minimization problem*: find  $\tilde{u} \in V(\Omega)$  such that

$$L(\tilde{u}) \leq L(u), \text{ for all } u \in V(\Omega)$$

- Introduce an additional *vector space*  $V_0(\Omega) \subseteq V(\Omega)$ , with homogeneous Dirichlet BC's  
 $V_0(\Omega) = \{u \in V(\Omega) \mid u(\mathbf{x}) = 0, \mathbf{x} \in \partial\Omega_D\}$
- *Extremum problem*: find  $\tilde{u} \in V(\Omega)$  such that the *Gâteaux variation* of  $L$  is zero at  $\tilde{u}$ , i.e.:

$$\left. \frac{d}{d\epsilon} L(\tilde{u} + \epsilon v) \right|_{\epsilon=0} = 0, \text{ for all } v \in V_0(\Omega)$$

- $V_0(\Omega)$  is called the *space of test functions*
- Rewrite the Gâteaux variation of  $L$  so that the du Bois-Reymond Lemma can be applied to derive an (Euler-Lagrange) equation for  $\tilde{u}$ . Solve the Euler-Lagrange equation on  $V(\Omega)$  to find  $\tilde{u}$

# Fundamental lemma of variational calculus

**Lemma (du Bois-Reymond):** Assume  $\Omega = (a, b)$ . Let  $V_0(\Omega)$  be a function space of sufficiently smooth functions that vanish on the boundary of  $\Omega$ . Assume  $f(x)$  is continuous on  $\overline{\Omega}$ , i.e.,  $f \in C(\Omega)$ . Suppose that

$$\int_a^b f(x)v(x) dx = 0, \text{ for all } v \in V_0(\Omega) \quad (*)$$

Then,

$$f(x) = 0, \quad x \in \overline{\Omega} \quad (**)$$

That is: a *weak* zero of  $f(x)$  (\*) established for all test functions from  $V_0(\Omega)$  is equivalent to the *strong* zero of  $f(x)$  (\*\*). Valid also in  $2D$  and  $3D$ .

## Example: shortest path in a plane

$\Omega = (a, b)$ .

$V(\Omega) = \{u(x) \in C^2(\Omega) \mid u(a) = u_a, u(b) = u_b\}$ .

$V_0(\Omega) = \{v(x) \in C_0^2(\Omega) \mid v(a) = v(b) = 0\}$ . Note,  $V_0(\Omega)$  is a vector space.

We want to minimize the functional:

$$L(u) = \int_a^b \sqrt{1 + (u')^2} dx \text{ over } V(\Omega)$$

We look for an extremum, where:  $\frac{d}{d\epsilon} L(\tilde{u} + \epsilon v)|_{\epsilon=0} = 0$  for all  $v \in V_0(\Omega)$ , i.e.,

$$\begin{aligned} \frac{d}{d\epsilon} L(\tilde{u} + \epsilon v) &= \frac{d}{d\epsilon} \int_a^b \sqrt{1 + ((\tilde{u} + \epsilon v)')^2} dx \\ &= \int_a^b \frac{d}{d\epsilon} \sqrt{1 + (\tilde{u}' + \epsilon v')^2} dx = \int_a^b \frac{(\tilde{u}' + \epsilon v') v'}{\sqrt{1 + (\tilde{u}' + \epsilon v')^2}} dx \\ \frac{d}{d\epsilon} L(\tilde{u} + \epsilon v) \Big|_{\epsilon=0} &= \int_a^b \frac{\tilde{u}' v'}{\sqrt{1 + (\tilde{u}')^2}} dx \end{aligned}$$

## Shortest path in a plane (continued)

$$\left. \frac{d}{d\epsilon} L(\tilde{u} + \epsilon v) \right|_{\epsilon=0} = 0, \quad \forall v \in V_0(\Omega)$$

$$\int_a^b \frac{\tilde{u}' v'}{\sqrt{1 + (\tilde{u}')^2}} dx = \left[ \begin{array}{ll} \phi = \frac{\tilde{u}'}{\sqrt{1 + (\tilde{u}')^2}}, & d\psi = v' dx \\ d\phi = \frac{\tilde{u}''}{(1 + (\tilde{u}')^2)^{3/2}} dx, & \psi = v \end{array} \right]$$

$$= \left[ \frac{v \tilde{u}'}{\sqrt{1 + (\tilde{u}')^2}} \right]_a^b - \int_a^b \frac{v \tilde{u}''}{(1 + (\tilde{u}')^2)^{3/2}} dx$$

$$= \int_a^b \left[ \frac{-\tilde{u}''}{(1 + (\tilde{u}')^2)^{3/2}} \right] v dx = 0, \quad \forall v \in V_0(\Omega)$$

From du Bois-Reymond's Lemma and since  $u \in V(\Omega)$ :

$$\frac{-\tilde{u}''}{(1 + (\tilde{u}')^2)^{3/2}} = 0, \quad \text{or} \quad \boxed{-\tilde{u}'' = 0, \quad x \in (a, b); \quad u(a) = u_a, \quad u(b) = u_b}$$

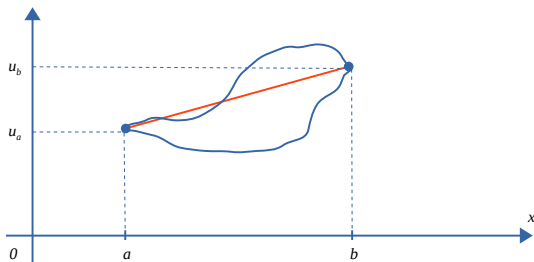
## Shortest path in a plane (continued)

Hence, extremum problem for the functional  $L(u)$  has been reduced to the boundary-value problem for the differential equation:

$$-\tilde{u}'' = 0, \quad \tilde{u}(a) = u_a, \quad \tilde{u}(b) = u_b$$

The solution is the straight line between  $(a, u_a)$  and  $(b, u_b)$ :

$$\tilde{u}(x) = u_a + \frac{u_b - u_a}{b - a}(x - a)$$



# Lagrange's equation in 1D

- $\Omega = (a, b)$
- $U(\Omega) = \{u(x) \in C^2(\Omega) \mid u(a) = u_a\}$
- $V(\Omega) = \{v(x) \in C^2(\Omega) \mid v(a) = 0\}$

Minimize

$$S(u) = \int_a^b \mathcal{L}(x, u, u') dx \text{ over } U(\Omega)$$

Additional assumptions:

- $\frac{\partial \mathcal{L}}{\partial u} \in C(\Omega)$
- $\frac{d}{dx} \left( \frac{\partial \mathcal{L}}{\partial u'} \right) \in C(\Omega)$
- $V_0(\Omega) = \{v(x) \in C^2(\Omega) \mid v(a) = v(b) = 0\}$

Note:  $V_0(\Omega) \subset V(\Omega)$ .

# Lagrange's equation in 1D (continued)

Let  $\tilde{u} \in U(\Omega)$ ,  $v \in V(\Omega)$ ,  $\epsilon \in \mathbb{R}$ :

$$\begin{aligned} \left. \frac{d}{d\epsilon} S(\tilde{u} + \epsilon v) \right|_{\epsilon=0} &= \frac{d}{d\epsilon} \int_a^b \mathcal{L}(x, \tilde{u} + \epsilon v, (\tilde{u} + \epsilon v)') dx \\ &= \int_a^b \frac{d}{d\epsilon} \mathcal{L}(x, \tilde{u} + \epsilon v, \tilde{u}' + \epsilon v') dx \\ &= \int_a^b \left[ \frac{\partial \mathcal{L}}{\partial \tilde{u}} \frac{d}{d\epsilon} (\tilde{u} + \epsilon v) + \frac{\partial \mathcal{L}}{\partial \tilde{u}'} \frac{d}{d\epsilon} (\tilde{u}' + \epsilon v') \right] dx \\ &= \int_a^b \left[ \frac{\partial \mathcal{L}}{\partial \tilde{u}} v + \frac{\partial \mathcal{L}}{\partial \tilde{u}'} v' \right] dx = \int_a^b \frac{\partial \mathcal{L}}{\partial \tilde{u}} v dx + \int_a^b \frac{\partial \mathcal{L}}{\partial \tilde{u}'} v' dx \\ &= \int_a^b \frac{\partial \mathcal{L}}{\partial \tilde{u}} v dx + \left[ \frac{\partial \mathcal{L}}{\partial \tilde{u}'} v \right]_a^b - \int_a^b \frac{d}{dx} \left( \frac{\partial \mathcal{L}}{\partial \tilde{u}'} \right) v dx \end{aligned}$$



## Lagrange's equation in 1D (continued)

Since  $v \in V(\Omega)$ ,  $v(a) = 0$ , and  $\frac{d}{d\epsilon} S(\tilde{u} + \epsilon v)|_{\epsilon=0} = 0 \Rightarrow$

$$\int_a^b \left[ \frac{\partial \mathcal{L}}{\partial \tilde{u}} - \frac{d}{dx} \left( \frac{\partial \mathcal{L}}{\partial \tilde{u}'} \right) \right] v \, dx = - \frac{\partial \mathcal{L}}{\partial \tilde{u}'} v \Big|_{x=b}, \quad \forall v \in V(\Omega) \quad (*)$$

If this holds for all  $v \in V(\Omega)$ , then it also holds for all  $v \in V_0(\Omega)$ , since  $V_0(\Omega) \subset V(\Omega)$ . Recall that  $v(b) = 0$  if  $v \in V_0(\Omega)$ . Further, by du Bois-Reymond's Lemma:

$$\frac{\partial \mathcal{L}}{\partial \tilde{u}} - \frac{d}{dx} \left( \frac{\partial \mathcal{L}}{\partial \tilde{u}'} \right) = 0 \quad (**)$$

On the other hand, if  $(**)$  holds, then from  $(*)$ :

$$\begin{aligned} - \frac{\partial \mathcal{L}}{\partial \tilde{u}'} v \Big|_{x=b} &= 0, \quad \forall v \in V(\Omega) \\ \text{i.e. } \frac{\partial \mathcal{L}}{\partial \tilde{u}'} \Big|_{x=b} &= 0, \quad \text{since } v(b) \neq 0 \end{aligned}$$

## Lagrange's equation in 1D (continued)

Hence, the minimization (extremum) problem for the action functional reduces to the Lagrange differential equation:

$$\frac{\partial \mathcal{L}}{\partial \tilde{u}} - \frac{d}{dx} \left( \frac{\partial \mathcal{L}}{\partial \tilde{u}'} \right) = 0, \quad x \in (a, b)$$

with the *essential* boundary condition:  $\tilde{u}(a) = u_a$ ,

and the *natural* boundary condition:  $\frac{\partial \mathcal{L}}{\partial \tilde{u}'}(b) = 0$

## Example: mass on a spring

Identify:  $x \rightarrow t$ ,  $\tilde{u} \rightarrow x$ ,  $\tilde{u}' \rightarrow x'$ , and the Lagrangian is

$$\mathcal{L}(t, x, x') = \frac{m(x')^2}{2} - \frac{kx^2}{2}$$

The Euler-Lagrange equation is:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial x} - \frac{d}{dt} \left( \frac{\partial \mathcal{L}}{\partial x'} \right) &= 0 \\ -kx - \frac{d}{dt} (mx') &= 0 \\ mx'' + kx &= 0, \quad t \in (a, b)\end{aligned}$$

Hence, the Lagrange equation further reduces to a linear second-order ODE (“oscillator” equation).

# Lagrangian functional for a self-adjoint operator

## Theorem:

- $\Omega$  – open bounded domain in  $\mathbb{R}^n$ ,  $n = 1, 2, 3$
- $V_0(\Omega)$  – space of functions on  $\Omega$  that are zero on  $\partial\Omega$
- $f \in C(\Omega)$
- $\mathcal{L}$  – positive self-adjoint linear differential operator on  $V_0(\Omega)$
- $u_0 \in V_0(\Omega)$  – the unique solution of the differential equation:

$$\mathcal{L}(u) = f$$

Then,  $u_0$  minimizes the functional:

$$L(u) = \int_{\Omega} \left[ \frac{1}{2} u \mathcal{L}(u) - fu \right] d\Omega$$

i.e.

$$L(u_0) \leq L(u), \quad \text{for all } u \in V_0(\Omega)$$

## Lagrangian for a self-adjoint operator (proof)

**Proof:** Let  $u, u_0 \in V_0(\Omega)$  and  $u \neq u_0$ . Also  $(u - u_0) \in V_0(\Omega)$  (homogeneous Dirichlet BC's). Since  $\mathcal{L}$  is positive on  $V_0(\Omega)$ ,

$$\frac{1}{2} \int_{\Omega} (u - u_0) \mathcal{L}(u - u_0) d\Omega > 0 \quad (\text{positive})$$

$$\frac{1}{2} \int_{\Omega} [u \mathcal{L} u - u \mathcal{L} u_0 - u_0 \mathcal{L} u + u_0 \mathcal{L} u_0] d\Omega \quad (\text{linear})$$

$$= \frac{1}{2} \int_{\Omega} [u \mathcal{L} u - u \mathcal{L} u_0 - u \mathcal{L} u_0 + u_0 \mathcal{L} u_0] d\Omega \quad (\text{self-adjoint})$$

$$= \frac{1}{2} \int_{\Omega} u \mathcal{L} u d\Omega - \int_{\Omega} u \mathcal{L} u_0 d\Omega + \frac{1}{2} \int_{\Omega} u_0 \mathcal{L} u_0 d\Omega$$

$$= \frac{1}{2} \int_{\Omega} u \mathcal{L} u d\Omega - \int_{\Omega} u \mathcal{L} u_0 d\Omega + \left(1 - \frac{1}{2}\right) \int_{\Omega} u_0 \mathcal{L} u_0 d\Omega$$

## Lagrangian for a self-adjoint operator (proof, continued)

$$\begin{aligned} \frac{1}{2} \int_{\Omega} (u - u_0) \mathcal{L}(u - u_0) d\Omega &= \\ &= \frac{1}{2} \int_{\Omega} u \mathcal{L}u d\Omega - \int_{\Omega} u \mathcal{L}u_0 d\Omega + \int_{\Omega} u_0 \mathcal{L}u_0 d\Omega - \frac{1}{2} \int_{\Omega} u_0 \mathcal{L}u_0 d\Omega \\ &= \frac{1}{2} \int_{\Omega} u \mathcal{L}u d\Omega - \int_{\Omega} u f d\Omega + \int_{\Omega} u_0 f d\Omega - \frac{1}{2} \int_{\Omega} u_0 \mathcal{L}u_0 d\Omega > 0 \quad (\mathcal{L}u_0 = f) \\ \frac{1}{2} \int_{\Omega} u \mathcal{L}u d\Omega - \int_{\Omega} u f d\Omega &> - \left[ \int_{\Omega} u_0 f d\Omega - \frac{1}{2} \int_{\Omega} u_0 \mathcal{L}u_0 d\Omega \right] \\ \int_{\Omega} \left[ \frac{1}{2} u \mathcal{L}u - u f \right] d\Omega &< \int_{\Omega} \left[ \frac{1}{2} u_0 \mathcal{L}u_0 - u_0 f \right] d\Omega, \quad \text{for all } u \in V_0(\Omega) \end{aligned}$$



# From functional to Poisson's equation

- $\Omega$  – open bounded domain in  $\mathbb{R}^n$ ,  $n = 1, 2, 3$
- $\partial\Omega = \partial\Omega_D \cup \partial\Omega_N$
- $k(\mathbf{x}) > 0$ ,  $\mathbf{x} \in \Omega$
- $f(\mathbf{x}) \in C(\Omega)$
- $g(\mathbf{x})$  is a function on  $\partial\Omega_D$
- $h(\mathbf{x})$  is a continuous function on  $\partial\Omega_N$
- $U(\Omega) = \{u \mid \nabla \cdot (k \nabla u) \in C(\Omega); u(\mathbf{x}) = g(\mathbf{x}), \mathbf{x} \in \partial\Omega_D\}$
- $V(\Omega) = \{v \mid \nabla \cdot (k \nabla v) \in C(\Omega); v(\mathbf{x}) = 0, \mathbf{x} \in \partial\Omega_D\}$
- $V_0(\Omega) = \{v \mid \nabla \cdot (k \nabla v) \in C(\Omega); v(\mathbf{x}) = 0, \mathbf{x} \in \partial\Omega\} \subset V(\Omega)$

Minimize the functional

$$F(u) = \int_{\Omega} \left[ \frac{1}{2} \nabla u \cdot (k \nabla u) - fu \right] d\Omega + \int_{\partial\Omega_N} hu d\Gamma \quad \text{over } U(\Omega)$$

## Minimization problem (continued)

Take  $u \in U(\Omega)$ ,  $v \in V(\Omega)$ ,  $\epsilon \in \mathbb{R}$ , and compute the variation:

$$\begin{aligned} F(u + \epsilon v) &= \\ &\int_{\Omega} \left[ \frac{1}{2} \nabla(u + \epsilon v) \cdot (k \nabla(u + \epsilon v) - f(u + \epsilon v)) \right] d\Omega + \int_{\partial\Omega_N} h(u + \epsilon v) d\Gamma \\ &= \int_{\Omega} \left[ \frac{1}{2} \nabla u \cdot (k \nabla u) + \frac{1}{2} \nabla u \cdot (k \nabla \epsilon v) + \frac{1}{2} \nabla \epsilon v \cdot (k \nabla u) \right. \\ &\quad \left. + \frac{1}{2} \nabla \epsilon v \cdot (k \nabla \epsilon v) - fu - \epsilon fv \right] d\Omega + \int_{\partial\Omega_N} (hu + \epsilon hv) d\Gamma \\ &= \int_{\Omega} \left[ \frac{1}{2} \nabla u \cdot (k \nabla u) - fu \right] d\Omega + \int_{\partial\Omega_N} hu d\Gamma \\ &\quad + \epsilon \int_{\Omega} [\nabla u \cdot (k \nabla v) - fv] d\Omega + \epsilon \int_{\partial\Omega_N} hv d\Gamma \\ &\quad + \frac{\epsilon^2}{2} \int_{\Omega} \nabla v \cdot (k \nabla v) d\Omega \end{aligned}$$



## Minimization problem (continued)

Compute the derivative at  $\epsilon = 0$  and transform integral to the form  $\int(\dots)v \, d\Omega$

$$\left. \frac{d}{d\epsilon} F(u + \epsilon v) \right|_{\epsilon=0} = 0$$

$$\int_{\Omega} [\nabla u \cdot (k \nabla v) - fv] \, d\Omega + \int_{\partial\Omega_N} hv \, d\Gamma = 0$$

$$\begin{bmatrix} \nabla \cdot (uk \nabla v) = \nabla u \cdot (k \nabla v) + u \nabla \cdot (k \nabla v) \\ \nabla u \cdot (k \nabla v) = \nabla \cdot (vk \nabla u) - v \nabla \cdot (k \nabla u) \\ \int_{\Omega} \nabla \cdot (vk \nabla u) \, d\Omega = \int_{\partial\Omega} (\mathbf{n} \cdot \nabla u) kv \, d\Gamma \end{bmatrix}$$

$$\int_{\partial\Omega} (\mathbf{n} \cdot \nabla u) kv \, d\Gamma + \int_{\Omega} [-\nabla \cdot (k \nabla u) - f] v \, d\Omega + \int_{\partial\Omega_N} hv \, d\Gamma = 0$$

$$\int_{\Omega} [-\nabla \cdot (k \nabla u) - f] v \, d\Omega + \int_{\partial\Omega_N} [(\mathbf{n} \cdot \nabla u)k + h] v \, d\Gamma = 0$$

## Minimization problem (continued)

$$\int_{\Omega} [-\nabla \cdot (k \nabla u) - f] v \, d\Omega + \int_{\partial\Omega_N} [(\mathbf{n} \cdot \nabla u)k + h] v \, d\Gamma = 0$$

Since this holds for all  $v \in V(\Omega)$  it also holds for all  $v \in V_0(\Omega)$ .

Therefore, by du Bois-Reymond's Lemma:

$$-\nabla \cdot (k \nabla u) = f, \quad \mathbf{x} \in \Omega \quad (*)$$

Thus, if we choose  $u \in U(\Omega)$  to satisfy this equation and the additional boundary condition

$$-(k \nabla u) \cdot \mathbf{n} = h, \quad \mathbf{x} \in \partial\Omega_N, \quad (**)$$

then we shall minimize the functional  $F[u]$  on  $U(\Omega)$ . The Dirichlet boundary condition  $u(\mathbf{x}) = g(x)$ ,  $\mathbf{x} \in \partial\Omega_D$  is not present in the functional, but is included in  $U(\Omega)$  and makes the solution of  $(*)$  unique.